

WEEKLY REPORT 6

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Week: 6 (Try enhancing visual quality of the image) Project Title: Super Resolution using Classical Interpolation Methods

Objective

This week's work was dedicated to enhance the visual quality of the reconstructed satellite images through post-processing, and begin for finalizing the things. The focus this week was on visual refinement, stability, and computational efficiency of the full SR pipeline (NEDI + IBP + Wavelet).

Work Done

a. Post-Enhancement and Refinement

- We tried adding a denoising and contrast-equalization step to reduce minor color noise from wavelet reconstruction.
 - Tried implementing bilateral filtering for noise suppression and also applied adaptive histogram equalization for restoring natural contrast.
- The enhanced SR images exhibited good clear road boundaries and more balanced tone.
- We were able to correct minor color distortion from previous week's outputs.

b. Parameter Optimization

- Conducted multiple test runs to tune:
 - IBP iterations: Optimal performance achieved at 4 – 5 iterations for 1024×1024 images.

- Wavelet gain factor: Stabilized at 1.4 for best trade-off between sharpness and smoothness.

Observations

- Post-processing significantly improved color balance and local contrast.
- IBP convergence was observed to be stabilized after the 4th iteration — further iterations improvement was negligible.
- We achieved PSNR ≈ 24 dB and SSIM ≈ 0.75 after enhancement.

Challenges

Large 4K images still do consume high memory in IBP loops. Amplification from wavelets occasionally produces light halo effects at strong edges.

Next Set Work

- Try integrating tile-based IBP to handle large-scale satellite images.
- Analyze comparative runtime vs quality for the final results.
- Assemble all visuals (HR/LR/SR comparisons, bar graphs, tables) and finalizing the methodology flow for final project presentation.